

II. SEXUAL ISOLATION IN THE MULLERI SUBGROUP

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INTRODUCTION

The mulleri subgroup of species belongs to the large repleta species group, of which more than fifty forms have been described. About sixteen species may be assigned to this subgroup, and eight of these have been selected for this study. These eight species, together with the strains used, are as follows:

- D. aldrichi* Patterson and Crow 1940; strain from Aldrich Farm, Austin, Texas.
- D. arizonensis* Patterson and Wheeler 1942; strain 928.5, Tucson, Arizona.
- D. buzzatii* Patterson and Wheeler 1942; strain 190, Cordoba, Argentina.
- D. hamatofila* Patterson and Wheeler 1942; strain from Aldrich Farm, Austin, Texas.
- D. mojavenensis* Patterson and Crow 1940; strain from Chocolate Mountains, California.
- D. mulleri* Sturtevant 1921; strain 1434.4, Aldrich Farm, Austin, Texas.
- D. peninsularis* Patterson and Wheeler 1942; strain 1148.7, Lake McKethan, Florida.
- D. ritae* Patterson and Wheeler 1942; strain from Huachuca Mountains, Arizona.

The members of this subgroup have certain characters in common, of which two may be mentioned. While the tergite pattern varies among the different species, yet there is a strong tendency for the abdominal bands to be broken up into spots, either in one or both sexes. The second character, which is constant for the subgroup, relates to the structure of the spermathecae. In most members of the genus *Drosophila* these paired organs serve as sperm receptacles, but in the mulleri series they are small and have slender stalks, and only rarely are the spermatozoa successful in gaining entrance to them. These organs have become modified into gland-like structures, each surrounded by a thick envelope from which secretions are discharged into its cavity.

In testing for sexual isolation in the mulleri subgroup, the writer has followed closely the experimental procedure used in the preceding article for the virilis group. It is, therefore, unnecessary to give an extended account of the methods employed. The life cycles of the mulleri forms are somewhat shorter than those for the virilis species, consequently, the flies were aged for only four days after emergence before making the crosses. The females were dissected after ninety-six hours exposure

to the males. In the sexual preference tests the same plan was followed as before; five females of each of the two species under test were placed in a vial with a single male of one of the species.

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FERTILITY TESTS AMONG THE EIGHT SPECIES OF THE MULLERI SUBGROUP

Three papers have been published which give the degree of fertility among some of the members of this subgroup. The first of these included but three species (Patterson and Crow 1940), while the second and third dealt with only five species (Crow 1942, Patterson 1942). At most this would involve only twenty different crosses, whereas, the eight species used in the present study give a total of fifty-six different combinations. Moreover, in the three published articles only a few of the exposed females were dissected, and the percentages of fertility were based mainly on the production of hybrids. In the present study we dissected at least one hundred females for each combination, and from these determined the exact number of females which had been inseminated, irrespective of whether or not hybrids had been produced. The number inseminated was obtained by the use of small mass matings of ten females and ten males each. The tests were carried out in this manner because mass matings often yield offspring in crosses in which pair matings fail to do so.

The results obtained from the fifty-six different combinations are given in Table 1. The first column shows the nature of the cross, the second indicates whether or not hybrids were obtained, the third gives the number of females dissected, and the fourth and last shows the number of females inseminated. The different crosses have been numbered from 1 to 56, for ease of reference, and these numbers appear in parentheses in the text. In thirty of the crosses none of the females had been inseminated, and, consequently, hybrids were not obtained. The number of females inseminated in the other twenty-six crosses varied from one to ninety-three. Ten of these yielded some sort of hybrids (1, 3, 5, 7, 9, 15, 21, 27, 31, 32), and sixteen failed to do so. The failure of the females of the latter group to produce offspring was due, in a large measure, to the deleterious effects of the insemination reaction in heterogamic matings (Patterson 1946). This reaction will be fully considered in the next article.

A further analysis of the data displayed in Table 1 may be made by referring individually to each of the eight species.

TABLE 1
Crosses between members of the *mulleri* subgroup

Crosses	Types of hybrids	♀ ♀ dissected	Inseminated
1. <i>mulleri</i> ♀ × <i>aldrichi</i> ♂	sterile ♀ ♀, ♂ ♂	100	32
2. <i>aldrichi</i> ♀ × <i>mulleri</i> ♂	none	100	18
3. <i>mulleri</i> ♀ × <i>arizonensis</i> ♂	sterile ♂ ♂	100	8
4. <i>arizonensis</i> ♀ × <i>mulleri</i> ♂	none	100	0
5. <i>mulleri</i> ♀ × <i>buzzatii</i> ♂	abnormal flies	100	2
6. <i>buzzatii</i> ♀ × <i>mulleri</i> ♂	none	101	0
7. <i>mulleri</i> ♀ × <i>hamatofila</i> ♂	sterile ♀ ♀, ♂ ♂	100	10
8. <i>hamatofila</i> ♀ × <i>mulleri</i> ♂	none	109	0
9. <i>mulleri</i> ♀ × <i>mojavensis</i> ♂	fert. ♀ ♀, st. ♂ ♂	100	21
10. <i>mojavensis</i> ♀ × <i>mulleri</i> ♂	none	105	0
11. <i>mulleri</i> ♀ × <i>peninsularis</i> ♂	none	104	0
12. <i>peninsularis</i> ♀ × <i>mulleri</i> ♂	none	104	0
13. <i>mulleri</i> ♀ × <i>ritae</i> ♂	none	110	0
14. <i>ritae</i> ♀ × <i>mulleri</i> ♂	none	104	0
15. <i>aldrichi</i> ♀ × <i>arizonensis</i> ♂	sterile ♀ ♀	100	26
16. <i>arizonensis</i> ♀ × <i>aldrichi</i> ♂	none	100	0
17. <i>aldrichi</i> ♀ × <i>buzzatii</i> ♂	none	100	0
18. <i>buzzatii</i> ♀ × <i>aldrichi</i> ♂	none	105	0
19. <i>aldrichi</i> ♀ × <i>hamatofila</i> ♂	none	100	11
20. <i>hamatofila</i> ♀ × <i>aldrichi</i> ♂	none	103	0
21. <i>aldrichi</i> ♀ × <i>mojavensis</i> ♂	sterile ♀ ♀	100	46
22. <i>mojavensis</i> ♀ × <i>aldrichi</i> ♂	none	100	15
23. <i>aldrichi</i> ♀ × <i>peninsularis</i> ♂	none	102	0
24. <i>peninsularis</i> ♀ × <i>aldrichi</i> ♂	none	105	0
25. <i>aldrichi</i> ♀ × <i>ritae</i> ♂	none	102	0
26. <i>ritae</i> ♀ × <i>aldrichi</i> ♂	none	106	0
27. <i>arizonensis</i> ♀ × <i>buzzatii</i> ♂	larvae only	106	1
28. <i>buzzatii</i> ♀ × <i>arizonensis</i> ♂	none	100	88
29. <i>arizonensis</i> ♀ × <i>hamatofila</i> ♂	none	100	21
30. <i>hamatofila</i> ♀ × <i>arizonensis</i> ♂	none	100	5
31. <i>arizonensis</i> ♀ × <i>mojavensis</i> ♂	fert. ♀ ♀, st. ♂ ♂	100	29
32. <i>mojavensis</i> ♀ × <i>arizonensis</i> ♂	fert. ♀ ♀, ♂ ♂	100	93
33. <i>arizonensis</i> ♀ × <i>peninsularis</i> ♂	none	102	0
34. <i>peninsularis</i> ♀ × <i>arizonensis</i> ♂	none	100	43
35. <i>arizonensis</i> ♀ × <i>ritae</i> ♂	none	101	0
36. <i>ritae</i> ♀ × <i>arizonensis</i> ♂	none	100	0
37. <i>buzzatii</i> ♀ × <i>hamatofila</i> ♂	none	100	74
38. <i>hamatofila</i> ♀ × <i>buzzatii</i> ♂	none	100	1
39. <i>buzzatii</i> ♀ × <i>mojavensis</i> ♂	none	100	18
40. <i>mojavensis</i> ♀ × <i>buzzatii</i> ♂	none	104	0
41. <i>buzzatii</i> ♀ × <i>peninsularis</i> ♂	none	103	0
42. <i>peninsularis</i> ♀ × <i>buzzatii</i> ♂	none	103	1
43. <i>buzzatii</i> ♀ × <i>ritae</i> ♂	none	100	0
44. <i>ritae</i> ♀ × <i>buzzatii</i> ♂	none	100	0
45. <i>hamatofila</i> ♀ × <i>mojavensis</i> ♂	none	100	19
46. <i>mojavensis</i> ♀ × <i>hamatofila</i> ♂	none	100	33
47. <i>hamatofila</i> ♀ × <i>peninsularis</i> ♂	none	103	0
48. <i>peninsularis</i> ♀ × <i>hamatofila</i> ♂	none	100	10
49. <i>hamatofila</i> ♀ × <i>ritae</i> ♂	none	100	1
50. <i>ritae</i> ♀ × <i>hamatofila</i> ♂	none	100	10
51. <i>mojavensis</i> ♀ × <i>peninsularis</i> ♂	none	103	0
52. <i>peninsularis</i> ♀ × <i>mojavensis</i> ♂	none	100	0
53. <i>mojavensis</i> ♀ × <i>ritae</i> ♂	none	110	0
54. <i>ritae</i> ♀ × <i>mojavensis</i> ♂	none	102	0
55. <i>peninsularis</i> ♀ × <i>ritae</i> ♂	none	103	0
56. <i>ritae</i> ♀ × <i>peninsularis</i> ♂	none	100	0

D. mulleri.—Females of this species were inseminated by the males of five of the other seven species. In four of the crosses (1, 3, 5, 7), such hybrids as were produced were sterile, and in the fifth cross (9) the F_1 males were also sterile, but at least some of the hybrid females were fertile in backcross. In the reciprocal cross, the *mulleri* males failed to fertilize any of the females of six other species, but did inseminate 18% of the *aldrichi* females without producing offspring (2). Several years ago the writer found another member of the subgroup, *D. longicornis*, the males of which fertilized *mulleri* females, producing a few sterile hybrids. The results from this test have been reported by Wharton (1944). So far as the exchange of genes is concerned, the results from these tests show that *D. mulleri* cannot do so except with one other species. In the *mulleri/mojavensis* cross, Crow (1942) has reported that in pair mating tests, 5% of the females produce offspring, and that the F_1 females are fertile in backcrosses to the parent types of males.

D. aldrichi.—Females of this species were inseminated by the males of four other forms, *mulleri*, *hamatofila*, *arizonensis*, and *mojavensis*. The first two crosses do not result in the production of offspring (2, 19) while the other two produce sterile females (15, 21). In the reciprocal crosses, *aldrichi* males inseminated *mulleri* females, producing sterile males and females (1), and also *mojavensis* females without the production of offspring (22). This species cannot, therefore, exchange genes with any other member of the subgroup to which it has been tested.

D. arizonensis.—The females of *arizonensis* were fertile to the males of two other species. The first successful cross was to *buzzatii* males and resulted in the production of larvae which died in mid-larval stages (27). The second cross to go was with *mojavensis* males, and yielded fertile females and sterile males (31). In the reciprocal crosses males of *arizonensis* inseminated the females of six other species (3, 15, 28, 30, 32, 34), but only three of these produced hybrids (3, 15, 32). The reciprocal crosses of *arizonensis* and *mojavensis* are two crosses in which an exchange of genes with other members of the subgroup is possible. The *mojavensis/arizonensis* cross was the most successful of the entire series, giving both fertile males and females.

D. buzzatii.—The females of this form were inseminated by the males of three other species, but none of these crosses produced hybrids. In the *buzzatii/arizonensis* cross (28), 88% of the females had been inseminated, and in the *buzzatii/hamatofila* cross (37) 74%. In the third cross, *buzzatii/mojavensis* (39), 18% of the females had been inseminated. In the reciprocal crosses, the males of *buzzatii* had inseminated four types of females, *mulleri*, *arizonensis*, *hamatofila*, and *peninsularis*. In the *mulleri/buzzatii* cross most of the resulting zygotes die in larval stages, and only a few reach maturity and are abnormal and sterile (5). The *arizonensis/buzzatii* cross gave a few lethal larvae (27). One female of *hamatofila* and one of *peninsularis* had been inseminated by *buzzatii*

males (38, 42). On the basis of these results this species cannot exchange genes with any other member of the subgroup.

D. hamatofila.—The females of this species were inseminated by the males of *arizonensis* (30), *buzzatii* (38), *mojavensis* (45), and *ritae* (49), but hybrids were not produced. Only *mulleri* females were fertile to *hamatofila* males, but the hybrids were sterile (7). In addition to this cross, the females of the other six species had been inseminated to varying degrees by *hamatofila* males (19, 29, 37, 46, 48, 50). This species is also not capable of gene transfer with other members of the subgroup.

D. mojavensis.—The females of this species were inseminated by *aldrichi* and *hamatofila* males, without the production of hybrids (22, 46), and fertile to *arizonensis* males, giving fertile males and females (32). The males were fertile to *mulleri*, *arizonensis*, and *aldrichi* females, producing fertile females and sterile males in the first two crosses and sterile females in the third (9, 31, 21). These males also inseminated *buzzatii* and *hamatofila* females, but without producing offspring (39, 45). There are, therefore, three possible chances of gene exchange with this form and other members of the subgroup (9, 31, 32).

D. peninsularis.—The males of *peninsularis* failed to inseminate a single female of any of the other seven species, but males of *arizonensis*, *buzzatii*, and *hamatofila* did inseminate some of its females (34, 42, 48). However, no offspring was produced, and gene exchange is not possible.

D. ritae.—Offspring were not produced in any of the fourteen possible crosses with this species, but ten females had been inseminated by *hamatofila* males (50), and one *hamatofila* female had been inseminated by a *ritae* male (49). Gene exchange between this species and other members of the subgroup is also not possible.

While some sort of hybrid is produced in ten of the fifty-six crosses, yet there are only three of these in which an exchange of genes is possible. These are the fertile F_1 females from the *mulleri*/*mojavensis* and *arizonensis*/*mojavensis* crosses, and the fertile F_1 males and females from the *mojavensis*/*arizonensis* cross. All other hybrids are sterile and some of them are abnormal, or else the zygotes die in larval or pupal stages. We have no information as to whether the flies which produce fertile hybrids ever mate in nature. So far as known, the distributional areas of *mulleri* and *mojavensis* are widely separated, but there is a possibility that those of *mojavensis* and *arizonensis* may overlap. The most extensive overlapping of distributional areas are those of *mulleri* and *aldrichi* (Patterson and Wagner 1943), but the F_1 flies from the only cross that goes, *mulleri* ♀ × *aldrichi* ♂ are completely sterile. The male hybrids from this cross have been found in nature and definitely identified, and since this is the only known instance in which it has been possible to identify *Drosophila* hybrids in nature, it seems worthwhile to present the evidence upon which this statement is based.

The hybrids from the *mulleri*/*aldrichi* cross resemble *mulleri* in most of their characters, but the testis of the male hybrid is rudimentary and

shows little coiling, and, as seen through the abdominal wall, appears in the form of a question mark (?). It is this character which enables one to identify these males in collections brought in from the field. Such males are identical with the hybrid males obtained in the laboratory from this cross. During July and August of 1940 and 1941 collections of *Drosophila* from the overlap zone of *mulleri* and *aldrichi* in Texas were examined for the presence of these hybrid males. These data are summarized in Table 2. All of the collections were made in the country, and none in towns or cities. In recording the place at which a given collection was made the nearest town is listed, but if no towns were near, it is recorded as from a Roadside Park (R. P., in the table). The number of *mulleri* and *aldrichi* males are given for each collection, followed by the percentage of *aldrichi* males in the male population of both species. The number of hybrid males is shown in the last column. Hybrid females must also be present in some of these collections, but they could not be detected with certainty.

TABLE 2

Per cent of hybrid males found in nature from the cross *mulleri* ♀ × *aldrichi* ♂

Date	Place, County	<i>mulleri</i> males	<i>aldrichi</i> males	Per cent of <i>aldrichi</i> ♂ ♂	Hybrid males
6/13/40	Selma, Bexar	34	188	84.6	1
7/16/40	R.P., Fayette	27	26	49.0	1
7/19/40	R.P., Guadalupe	115	147	56.1	4
8/2/40	Selma, Bexar	100	99	49.7	1
8/2/40	R.P.(A), Comal	279	340	54.8	2
8/2/40	Cactus, LaSalle	5	11	68.7	1
8/2/40	Devine, Medina	24	16	40.0	1
8/9/40	R.P., Williamson	56	212	79.1	9
8/9/40	Troy, Bell	92	50	35.2	2
8/9/40	R.P.(A), Bell	153	148	49.1	3
8/9/40	R.P.(B), Bell	73	41	35.1	1
7/27/41	R.P., Travis	14	36	72.0	3
7/27/41	R.P.(A), Williamson	16	15	48.4	1
6/7/40	San Antonio, Bexar	19	6	24.0	0
8/2/40	R.P., Hays	38	15	28.3	0
8/2/40	R.P.(B), Comal	80	32	28.5	0
8/2/40	R.P., Guadalupe	177	64	26.5	0
8/2/40	Laredo, Webb	19	1	5.0	0
8/9/40	R.P., McLennan	52	3	5.4	0
8/12/40	Franklin, Robertson	190	72	27.4	0
8/12/40	Palestine, Anderson	60	15	20.0	0
7/28/41	R.P.(B), Williamson	68	16	19.0	0

There are twenty-two collections listed in the table. In the first thirteen collections there are 2,317 normal and thirty hybrid males. In the last nine collections there are 927 normal males and no hybrids, although one would expect about twelve such hybrids with this number of normal males, that is, if the populations were alike. Evidently, the size of the collection does not determine the presence or absence of hybrids. For example, a single hybrid male was present in each of the collections from Selma (6/13/40) and Cactus (6/2/40), and yet the first contained 222

normal males and the latter only 16. In so far as the samples in the table are concerned, the presence or absence of hybrids must depend on the proportion of *aldrichi* males in the male population in the two species. In all collections in which these males exceed one-third of this population hybrids are present (first thirteen), but they are absent where it is less than a third (last nine).

SEXUAL PREFERENCE TESTS

As pointed out above, thirty of the fifty-six crosses listed in Table 1 gave no offspring, and none of the females had been inseminated. If in heterogamic tests the male, without choice of mate, does not inseminate any of the females, it is most unlikely that he would do so where a choice is possible. There would, therefore, be no point in carrying out sexual preference tests on these thirty combinations, for the results already indicate that sexual isolation is complete. There were four crosses in which a single female was found to have been inseminated. Such cases cannot be attributed to experimental error, because in each instance the vagina of the dissected female had an old reaction mass of the heterogamic type. There would also not be much point in running sexual preference tests on crosses in which a single female had been inseminated in heterogamic matings. We have, however, tested two of the four such crosses listed in the table. This makes a total of twenty-four crosses to be tested for sexual preference by the multiple choice technique. The results obtained are given in Table 3.

TABLE 3

Number of females dissected (*n*) and per cent (%) inseminated in various interspecific crosses of the *mulleri* subgroup

Females	Males	Homogamic		Heterogamic		χ^2	Iso- lation index
		<i>n</i>	%	<i>n</i>	%		
1. <i>mulleri</i> , <i>aldrichi</i>	<i>mulleri</i>	100	78	100	6	118.4	0.85
2. <i>aldrichi</i> , <i>mulleri</i>	<i>aldrichi</i>	100	63	100	4	78.1	0.88
3. <i>arizonensis</i> , <i>mulleri</i>	<i>arizonensis</i>	100	61	100	5	70.9	0.85
4. <i>buzzatii</i> , <i>mulleri</i>	<i>buzzatii</i>	100	92	100	0	170.4	1.00
5. <i>hamatofila</i> , <i>mulleri</i>	<i>hamatofila</i>	100	77	100	3	114.1	0.92
6. <i>mojavensis</i> , <i>mulleri</i>	<i>mojavensis</i>	100	75	100	11	83.6	0.74
7. <i>arizonensis</i> , <i>aldrichi</i>	<i>arizonensis</i>	100	51	100	10	39.6	0.67
8. <i>hamatofila</i> , <i>aldrichi</i>	<i>hamatofila</i>	100	75	100	3	108.9	0.92
9. <i>mojavensis</i> , <i>aldrichi</i>	<i>mojavensis</i>	100	83	100	20	88.6	0.61
10. <i>aldrichi</i> , <i>mojavensis</i>	<i>aldrichi</i>	100	73	100	9	84.7	0.78
11. <i>buzzatii</i> , <i>arizonensis</i>	<i>buzzatii</i>	100	90	100	0	163.6	1.00
12. <i>arizonensis</i> , <i>buzzatii</i>	<i>arizonensis</i>	100	16	100	58	36.9	—0.56
13. <i>hamatofila</i> , <i>arizonensis</i>	<i>hamatofila</i>	100	63	100	6	74.0	0.82
14. <i>arizonensis</i> , <i>hamatofila</i>	<i>arizonensis</i>	100	74	100	2	110.0	0.94
15. <i>mojavensis</i> , <i>arizonensis</i>	<i>mojavensis</i>	100	89	100	10	124.8	0.79
16. <i>arizonensis</i> , <i>mojavensis</i>	<i>arizonensis</i>	100	60	100	61	0.21	—0.01
17. <i>arizonensis</i> , <i>peninsularis</i>	<i>arizonensis</i>	100	60	100	9	57.5	0.74
18. <i>hamatofila</i> , <i>buzzatii</i>	<i>hamatofila</i>	100	55	100	55	0.0	0.00
19. <i>mojavensis</i> , <i>buzzatii</i>	<i>mojavensis</i>	100	92	100	7	144.5	0.85
20. <i>buzzatii</i> , <i>peninsularis</i>	<i>buzzatii</i>	100	86	100	0	150.9	1.00
21. <i>mojavensis</i> , <i>hamatofila</i>	<i>mojavensis</i>	100	80	100	5	115.1	0.88
22. <i>hamatofila</i> , <i>mojavensis</i>	<i>hamatofila</i>	100	64	100	13	54.9	0.66
23. <i>hamatofila</i> , <i>peninsularis</i>	<i>hamatofila</i>	100	72	100	9	82.3	0.77
24. <i>hamatofila</i> , <i>ritae</i>	<i>hamatofila</i>	100	78	100	0	127.8	1.00

The χ^2 values and the isolation indices for a large majority of these crosses show that the differences between the homogamic and heterogamic percentages are highly significant, and any extended comments on them would be superfluous. In four of the crosses (4, 11, 20, 24, Table 3) no heterogamic matings were obtained, although in the corresponding interspecific crosses, in which a choice of mate was not possible, some inseminations were found in each case (5, 27, 42, 49, Table 1). In two of these crosses a few hybrids were produced. In the *mulleri/buzzatii* cross (5) a few abnormal flies were obtained, and in the *arizonensis/buzzatii* cross (27) only inviable larvae were found. Of the other two crosses, the *peninsularis/buzzatii* cross (42) yielded a single inseminated female, and in the *ritae/hamatofla* cross (50) ten inseminated females were found. However, the isolation indices of the sexual preference tests indicate that sexual isolation is complete for these four combinations.

Another point of interest is that *arizonensis* males prefer *buzzatii* females to their own (12). The χ^2 value of 36.9 and the negative isolation index of -0.56 both indicate a very significant difference between the homogamic and heterogamic matings. A negative isolation index is not unusual. Dobzhansky *et al.* have reported several such cases in their studies on different forms, and in the preceding article we found one case between *americana* and *texana*, as well as several others between geographic strains of *D. virilis*. The males of *arizonensis* also show a preference for *mojavensis* females, but the difference of percentages is without significance. Finally *hamatofla* males inseminated *buzzatii* females and their own with equal frequency.

The extent to which sexual isolation is responsible for preventing the exchange of genes between the different members of the *mulleri* subgroup can be emphasized by reference to the data listed in Table 1. A total of 5,690 females of the fifty-six crosses were dissected, and it was found that 636 had been inseminated. This gives a percentage of slightly over eleven. Therefore, approximately 89% of the opportunities for gene exchange were prevented by sexual isolation. It is well recognized that one cannot measure exactly the effect of sexual isolation in nature. However, these experiments demonstrate the qualitative effectiveness of sexual isolation, and also the quantitative effectiveness within the limits of experimental procedure.

As one would expect, sexual isolation is more effective in preventing gene exchange in some crosses than in others. There were thirty combinations in which no inseminations were found among 3,081 dissected females. Moreover, the different species vary considerably in their reluctance to mate with other forms. In Table 4 are summarized the data which shows these differences. It shows for each species the number of crosses involved, the number of combinations in which its females had been inseminated by other males, and the number of combinations in which its males had inseminated females of other species. The per cents

given are based on two hundred dissected females for each individual cross.

TABLE 4

Species	Number of crosses	Females		Males	
		combinations	% inseminated	combinations	♀ %inseminated
hamatofila.....	11	4	4.1	7	18.4
arizonensis.....	9	3	11.1	6	34.8
mojavensis.....	8	3	37.3	5	18.6
buzzatii.....	7	3	51.6	4	0.5
aldrichi.....	6	4	17.5	2	15.0
mulleri.....	6	5	9.6	1	12.0
peninsularis.....	3	3	12.0	0	0.0
ritae.....	2	1	5.0	1	0.5

Drosophila hamatofila leads the list in the number of crosses in which it was involved. Its females were inseminated by the males of four other species, and its males inseminated the females of seven different species. The highest per cent of females inseminated by other males was *buzzatii* females, with a percentage of 51.6 in three combinations. But the per cent of females inseminated by *buzzatii* males was one of the two lowest, with a percentage of only 0.5 in four combinations. The six combinations with *arizonensis* males gave the highest per cent (34.8) for inseminations of alien females.

The results obtained in this series of experiments make it clear that sexual isolation is a real barrier to gene exchange between the members of the *mulleri* subgroup. It represents one of the most important factors in the evolutionary divergence of this group of species. This conclusion is supported by the work of other investigators on several other different groups of *Drosophila* species. It is not, however, the only barrier to gene transfer. There are several other isolating mechanisms operating in the *mulleri* subgroup, including the effects of the insemination reaction. All of these will be considered in the next article.

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- Other references to the literature on sexual isolation will be found at the end of Article I.